Full volume from shared-radius certificates

Original source and complete geometry

Fixed certified simple-white plat; prescribed knot-producing parity.
Exact shared-face coefficient, marked zeros, both level parities.

$$\lim_{N\to\infty} \frac{\log Z_{D,N}}{N} \quad = \quad \max_{q\in\mathcal{Q}_D} \sum_e h(q_e) \quad = \quad \frac{\mathrm{Vol}(Y_D)}{2\pi}$$

Entire CROSS filling period: proved

Nonzero geometric saddle; all original Fourier charges and
whole-domain endpoint currents controlled.

$$\lim_{N\to\infty} \frac{2\pi}{N} \log |I_{\mathrm{cross},N}| \quad = \quad \mathrm{Vol}(S^3 \ \setminus \ K_D(\mathbf{p}))$$

At most $2R$ shared-radius certificates

Choose one LEFT/RIGHT region; bound every other region in full.
Each test uses one common face vector, with protected zeros.

$C_{e,s}(L^{e,s}) \quad < \quad \Phi_D$ for every surviving anchor.

Complete-packing balance implies this condition.
Six rational witnesses certify a marked diagram where balance fails.

Full original invariant: proved under the certificate

The LEFT/RIGHT remainder has a strict global exponential gap.

$$\lim_{N\to\infty} \frac{2\pi}{N} \log |K_N(K_D(\mathbf{p}))| \quad = \quad \mathrm{Vol}(S^3 \ \setminus \ K_D(\mathbf{p}))$$

Every independently long fixed signed twist vector;
threshold depends on D. Twists are fixed before $N \ \to \ \infty$.

Proved family at EVERY fixed bridge width $k \ \geq \ 2$

Three-layer core plus an arbitrary marked unit-chain prefix.
$3k \ - \ 3 \ + \ 2m$ independent regions, for every fixed $m \ \geq \ 0$.

Exponents divisible by 3 give minimal bridge number exactly k.

Open: extend to general plat diagrams

The finite shared-radius theorem applies wherever its tests hold.
Their validity for every eligible plat remains to be proved.